

Track Model

Technical Specification

Open Data

Issue record

Issue	Date	Comments
V1.0	Dec 2022	Initial draft
V2.0	June 2023	Revised for release
V2.1	July 2023	Revised for new Ordnance Survey licence number
V2.2	June 2024	Revised for new ESRI Shapefile names
V3.0	October 2025	Revised to compliment the release of Network Rail's Geospatial Track Centre Line data on RDM
V3.1	April 2026	Revised for new Ordnance Survey copyright date
V3.2	June 2026	Revised to compliment the release of Network Rail's Track Gradient and Calibrated Point Measure Values data on RDM

Purpose of this specification

This is the Technical Specification for Network Rail's Track Model that will be made available from Network Rail on the Rail Data Marketplace (RDM).

This data may only be used under the [Open Government Licence](#) full details of which are available via the [Rail Data Marketplace website](#).

What this specification does not detail is the sources of information that have been used to compile this data although sources do include Ordnance Survey data. Ordnance Survey have been consulted prior to the data's release and have given their permission for the model to be the released under the Open Data terms and conditions and the appropriate copyright statements must be used alongside this data.

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1 Introduction

1.1 Overview

This document describes the Track Model data, named *NWR_TrackModel* on Rail Data Marketplace. It is a set of geospatial data that depicts the rail network. Within the data is a link-node model representing track centre lines, which is attributed with ownership information. The track centre line approximates the track alignment equidistant between the rail heads. In addition to the link-node model, the data also contains the locations for waymarks (approximate position of mileposts) as well the Engineers Line Reference (ELR) lines.

It is recommended that this documentation is read in conjunction with the Network Rail Geospatial Track Centre Line Technical Specification that is also available on the RDM.

This data is managed by Network Rail and is made available as open data on the Rail Data Marketplace. The data is refreshed and published every four weeks.

1.2 Purpose

The purpose of this document is to support users of the data, providing a high-level overview of what is included within the data and some terminology that is bespoke to the Railway. It defines what geospatial data layers are available, the schema of that data, the file formats, and the coordinate system.

1.3 Related Data Published on Rail Data Marketplace

Network Rail's Track Model dataset is undergoing a nationwide update to improve the geospatial accuracy of the link-node model (track centre line).

The Geospatial Track Centre Line (GTCL) dataset, first released on the Rail Data Marketplace in 2025, contains enhanced geospatial positioning of the track centre line data, derived from train mounted lidar and aerial imagery. While the GTCL data is geospatially more accurate, it does not contain the same attributes as the Track Model dataset.

Network Rail are in the process of merging the GTCL with the Track Model data at a pace controlled by data validation and system integration. The GTCL dataset will contain attributes which will inform end users whether a link has been synched into the Track Model dataset. The GTCL data will be refreshed and published every four weeks in line with this Track Model dataset.

If new track alignment has been generated through planned engineering works since the creation of the GTCL data, the new alignment will only be reflected in the Track Model dataset through business-as-usual processes.

The **Track Model dataset remains the master** dataset for ELR, track and mileage attribution within the *NWR_TrackCentreLines.shp* dataset.

1.4 Data Formats

The four components of the Track Model have been created as Esri shapefiles (.shp)

1.5 Coverage

The Track Model has national coverage of the rail network for Great Britain (England, Wales, and Scotland). Any tracks that do not have a history of having been owned and or maintained by Network Rail are not likely to be found within the data.

1.6 Credits

This has embedded Ordnance Survey (OS) data, with permission from OS and under the [Open Government Licence](#) Network Rail is able to release the data. The following copyright statement must be displayed whenever this data is used, either electronically or printed.

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2 Track Model Creation for Release as Open Data

The data being released as open data has been derived from the master Track Model data contained within a geodatabase on the corporate network; its primary function is to support Network Rail's monitoring fleet activities.

The information pertaining to ownership is not mastered within the Track Model, a new process has been developed to enable this information to be attributed to facilitate the release of the data. Ownership information is only available at track level within the *NWR_TrackCentreLines.shp* file, providing indicative rather than absolute information.

The data is being released on a four weekly update frequency under the [Open Government Licence](#). The data is for information only and must not be used to derive any safety critical decisions.

3 Spatial Data Extents and Coordinate Systems

Coverage: National

Projection: British National Grid

Maximum Extent:

North: 967997

East: 669557

South: 30543

West: 147517

4 Track Model Data and Schema

The Track Model contains four components; this section details these components and the schema for each of them in turn. All files are released as national coverage as an ESRI Shapefile.

4.1 Links

The *NWR_TrackCentreLines.shp* data contains an approximate geographic centre line representation of the rail network at track level. The links start/end for the following reasons: convergence or divergence of the tracks, buffer end stops or a change in key attribution used for the railway ELR or Track Identification (TrackID) information. A high-level explanation of TrackID can be found in the Appendix.

The ownership information is attributed to this dataset only. The ownership field provides a percentage value relating to the length of track identified as being in one of the following three categories: Network Rail, Third Party or Unclassified. More information is found in the schema for this data. This is displayed as a concatenated string, if the percentage is '0' in any classification all three categories will still be present.

With a confidence level of 95% of the rail network that is maintained by Network Rail the track centre line data has a geospatial accuracy of within +/- 1.5m. Higher accuracy data that has been transferred from the enhanced Geospatial Track Centre Line to the Track Model may have an accuracy up to +/- 0.25 metres, with a 95% confidence level.

The data type is polyline.

4.1.1 NWR_TrackCentreLines.shp Schema

Attribute Name	Data Type	Description
FID*	Object ID	Unique identifier generated when the shapefile is created.
Shape*	Geometry	Polyline – Represents the approximate centre line of the railway track.
ASSET_ID	Text - 12	Unique ID which is consistent and maintained as far as possible. ASSET_ID's are retired/introduced when track layouts change significantly. This always starts with the prefix of '9001'.
ELR	Text - 4	Engineers Line Reference, a railway term used to refer to a section of track covering multiple lines.
TRACK_ID	Text - 4	A four-digit code used within the railway to identify each section of track which indicates speed, direction and type of use.
START	Double	The start calibrated mileage value for the link. The value is always in mile.yards irrespective of the unit of measurement within the corresponding <i>NWR_Waymarks.shp</i> data.
END	Double	The end calibrated mileage value for the link. The value is always in mile.yards irrespective of the unit of measurement within the corresponding <i>NWR_Waymarks.shp</i> data.
VERSION	Double	This is the version number linked to the ASSET_ID which is increased each time the data is amended in the main system, this is either its attribution or geometry.
OWNER	Text - 100	A concatenated comma separated data field which shows the percentage of track for each link by distance, for which the owner is identified as: 'NETWORK RAIL', 'THIRD PARTY' or 'UNCLASSIFIED'. Unclassified mostly refers to third party data where all information is not available or where data is being updated and is awaiting confirmation from the business. The percentage is to the nearest whole number and is rounded up to 1% to ensure that the existence of different information is represented. For this reason, the percentage values should only be used as an approximation and not as a definitive convertible distance length.
EXTRACTED	Date	The date the <i>NWR_TrackCentreLines.shp</i> and corresponding information was extracted from the master data to create the Open Data

*Attribution added when ESRI Shapefile was created

4.2 Track Nodes

The *NWR_TrackNodes.shp* data, derived from the *NWR_TrackCentreLines.shp* dataset, occur at the start and or end of a link, with a minimum of one connected link and a maximum of four connected links as detailed in the *NWR_TrackNodes.shp* Schema.

The data type is point.

4.2.1 NWR_TrackNodes.shp Schema

Attribute Name	Data Type	Description
FID*	Object ID	Unique identifier generated when the shapefile is created.
Shape*	Geometry	Point – Represent the start and end of every <i>NWR_TrackCentreLines.shp</i>
VALANCY	Double	The number of links that connect to the point, this value can be between 1 and 4: '1' = Tracks start/end or no connecting data held '2' = Pseudo Node – no change in direction is not possible, attributional change only '3' = Directional change possible '4' = Centre of diamond crossing no directional change possible at this location although attributional changes may occur.
VERSION	Double	This is the version number linked to the ASSET_ID which is increased each time the data is amended in the main system, either its attribution or geometry.
EXTRACTED	Date	The date the <i>NWR_TrackNodes.shp</i> were extracted from the master data to create the Open Data
FID*	Object ID	Unique identifier generated when the shapefile is created.

*Attribution added when ESRI Shapefile was created

4.3 Engineers Line Reference

The *NWR_ELRs.shp* is a geographical representation of the model at route level. This is a virtual line which has been generated by creating a best fit between the main lines of each Engineer's Line Reference (ELR) and as such is generalised and not a seamless linear feature.

The reference line extends beyond the usable extent of the tracks to varying lengths to allow for assets associated with any given ELR to be visualised within Network Rail.

There is only one polyline for each ELR. Where there are two sections of an ELR that are not connected on the ground a straight line is used to connect these sections and does not represent the historical track location. Engineers Reference Line information is visible with no associated tracks which are there for historical purposes.

The data type is polyline

4.3.1 NWR_ELRs.shp Schema

Attribute Name	Data Type	Description
FID*	Object ID	Unique identifier generated when the shapefile is created.
Shape*	Geometry	Polyline – Represent the general location of the ELR
ASSET_ID	Text - 12	Unique ID which is consistent and maintained as far as possible. Each ELR has only one ASSET_ID. This always starts with the prefix of '9003'.
ELR	Text - 4	Engineers Line Reference, a railway term used to refer to a section of track covering multiple lines.
START	Double	The start calibrated mileage value for the ELR The value is always in mile.yards irrespective of the unit of measurement within the corresponding <i>NWR_Waymarks.shp</i> data.
END	Double	The end calibrated mileage value for the ELR. The value is always in mile.yards irrespective of the unit of measurement within the corresponding <i>NWR_Waymarks.shp</i> data.
VERSION	Double	This is the version number linked to the ASSET_ID which is increased each time the data is amended in the main system, this is either its attribution or geometry.
EXTRACTED	Date	The date the <i>NWR_ELRs.shp</i> were extracted from the master data to create the Open Data

*Attribution added when ESRI Shapefile was created

4.4 Waymarks

The *NWR_Waymarks.shp* data contains the approximate location of milepost for the national network as well additional waymarks which have been added to support business needs. For user awareness, it is known and understood that the waymarks are not always a precise measurement of 440 yards between each post where the line is measured in miles.

The data is also used to calibrate the linear referencing system for asset location.

The data type is point

4.4.1 NWR_Waymarks.shp Schema

Attribute Name	Data Type	Description
FID*	Object ID	Unique identifier generated when the shapefile is created.
Shape*	Geometry	Point – Represents the approximate location of a milepost or is a point added to support business needs.
ASSET_ID	Text - 12	Unique ID which is consistent and maintained as far as possible. ASSET_ID's are retired/introduced when track layouts change significantly. This always starts with the prefix of '9004'.
ELR	Text - 4	Engineers Line Reference, a railway term used to refer to a section of track covering multiple lines.
UNIT	Text - 1	This denotes the master unit of measurement for the ELR: 'K' = Kilometres 'M' = Miles
VALUE	Double	This is the locational value in either miles or kilometres of point
VERSION	Double	This is the version number linked to the ASSET_ID which is increased each time the data is amended in the main system, this is either its attribution or geometry.
EXTRACTED	Date	The date the <i>NWR_Waymarks.shp</i> and corresponding information was extracted from the master data to create the Open Data

*Attribution added when ESRI Shapefile was created

Appendix A

High level Track ID/Codes Description Overview

Track Name	Track ID
UP MAIN FAST	1100
UP SLOW	1200
UP GOODS	1300
UP SINGLE	1400
UP LOOP	1500
UP TERMINAL	1600
UP CROSSOVER	1700
UP OTHER/ENGINE	1800
UP SIDING	1900
DOWN MAIN FAST	2100
DOWN SLOW	2200
DOWN GOODS	2300
DOWN SINGLE	2400
DOWN LOOP	2500
DOWN TERMINAL	2600
DOWN CROSSOVER	2700
DOWN OTHER/ENGINE	2800
DOWN SIDING	2900
REVERSIBLE/BI-DIRECTIONAL MAIN FAST	3100
REVERSIBLE/BI-DIRECTIONAL SLOW	3200
REVERSIBLE/BI-DIRECTIONAL GOODS	3300
REVERSIBLE/BI-DIRECTIONAL SINGLE	3400
REVERSIBLE/BI-DIRECTIONAL LOOP	3500
REVERSIBLE/BI-DIRECTIONAL TERMINAL	3600
REVERSIBLE/BI-DIRECTIONAL CROSSOVER	3700
REVERSIBLE/BI-DIRECTIONAL OTHER/ENGINE	3800
REVERSIBLE/BI-DIRECTIONAL SIDING	3900